Adam Gillespie

3rd Year Mechanical & Manufacturing Engineering Student

Location: Ireland **GitHub**: https://github.com/Adam76e54

Skills Profile

Programming	C++, MATLAB, Rust, Python.
Embedded Systems	ISRs, PID control, FreeRTOS, networking APIs
CAD	Revit (BIM software), Onshape, SOLIDWORKS

Education & Academic Awards

2024 – Present **Mechanical & Manufacturing Engineering, MAI**
Trinity College Dublin

First Year Average: 78%

Second Year Average: 80%

Foundation Scholarship: elected following first-class performance in Scholarship examinations. Described by Trinity as "the most prestigious undergraduate award in the country".

Entrance Exhibition Award: awarded for Leaving Certificate performance.

2022 **Leaving Certificate**
Sancta Maria College, Co. Mayo

Achieved 613 CAO points

Projects

2026 **Line-Following Robot | Arduino R4, C++**

A small buggy which followed lines, stopped for obstacles, followed specified speed profiles, and could be controlled remotely through a Wi-Fi network and a custom GUI. Specific contributions included:

- Implemented PID speed control and wheel-speed matching, alongside interrupt-driven line following.
- Developed sensor drivers and a custom FIFO buffer for processing incoming network packets.
- Integrated motor-drive and wheel-encoder circuitry, using RC filtering to condition encoder signals.
- Used FreeRTOS to schedule sensing, control and communication tasks.

2026 **Machine-Vision Tool Tracking System | Raspberry Pi 4, Python**

As part of a team on a two-day hackathon:

- Developed a prototype camera-based system to detect hammers using a fine-tuned version of MobileNetV2 and a headless Raspberry Pi.
- Used OpenCV to capture webcam images.
- Built and soldered a limit switch circuit to detect an open toolbox drawer.

2025 **Custom Data Structures | C++**

Implemented custom data structures such as:

- A double-ended priority queue based on a two-layered binary heap that allowed constant-time access to both minimum and maximum objects.
- A custom open-addressed hash table.
- Implemented basic graph algorithms like Dijkstra's and Kruskal's for building a 3D maze generator and solver.

2026 – Present **Wave-Basin Testing**

Trinity Floating Wind — recently joined the wave-basin team developing a test facility for prototype floating wind turbines.

Previous Employment

2022 – 2023 And 2026 **Kitchen Porter**
Savoir Fare, Co. Mayo

- Washed dishes, made bread, made dishes like salads, soups, mussels.

2024 – 2025 **Barista**
The Creel Deli, Co. Mayo

- Served customers, completed end-of-day cash-ups, and managed stock ordering.
- Supervised and trained junior staff while coordinating a busy service floor.